Aim: A system has RAM which can store four pages simultaneously to satisf* page requests. Write a Program to calculate percentage page hit and page fault if he system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

os 8.cpp - Code::Blocks 25.03

```c
#include <stdio.h>

int main() {
    int frames[3], time[3];
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;
    int i, j, pos, min;
    int faults = 0, hits = 0;
    int counter = 0;

    // Initialize frames and time
    for(i = 0; i < 3; i++) {
        frames[i] = -1;
        time[i] = 0;
    }

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check for hit
        for(j = 0; j < 3; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }
```

*os 8.cpp - Code::Blocks 25.03

```c
        if(found == 0) {
            faults++;

            // Find least recently used page
            pos = 0;
            min = time[0];

            for(j = 1; j < 3; j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            counter++;
            time[pos] = counter;
        }
    }

    // Output
    printf("Page Faults = %d\n", faults);
    printf("Page Hits = %d\n", hits);
    printf("Hit Ratio = %.2f%%\n", (float)hits/n * 100);
    printf("Fault Ratio = %.2f%%\n", (float)faults/n * 100);

    return 0;
}
```

```
Page Faults = 10
Page Hits = 2
Hit Ratio = 16.67%
Fault Ratio = 83.33%

Process returned 0 (0x0)   execution time : 0.750 s
Press any key to continue.
```